Digital Savings Account Opening

Business Analysis Project (Banking)

Tushar Jain

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Self-made portfolio project. ABC Bank Ltd is a fictional bank, and the scenario, names and figures are my own. The KYC steps follow how digital account opening works in India under RBI's KYC and Video KYC (V-CIP) rules.

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1. Understanding the need

Opening a savings account at most traditional banks still means going to a branch, filling a long paper form, giving photocopies of PAN and Aadhaar and waiting 2 to 3 working days. Fintechs and digital banks open accounts in minutes, so banks that are still branch-based lose younger customers.

RBI now allows banks to do full KYC remotely: PAN can be checked with NSDL, Aadhaar details can be fetched through OTP eKYC, and Video KYC (V-CIP) with a bank officer replaces the in-person check. This project moves ABC Bank's savings account opening to the mobile app and website using these steps.

Requirement gathering

I prepared a questionnaire for each of the 4 modules and met the people who own that area. Main points:

Module	Who I met	What they told me
Onboarding & documents	Product Manager, IT Head	Name, DOB, PAN, Aadhaar, address, occupation, income needed. PAN image read by OCR and customer can edit it. Blurred image = re-upload, max 3 tries. Save and continue within 7 days.
KYC & AML	Head of Compliance, AML team	PAN check with NSDL, Aadhaar OTP eKYC with consent, CKYC search, duplicate check. Video KYC needed for full KYC. AML score: low continue, medium flag, high or sanctions match stop.
CBS account creation	CBS team lead, Product Manager	Create account only after all checks pass. Use existing CIF if the customer already banks with us. Products SAV001/002, SAV003 for 60+. CBS fails = retry twice, then Ops queue.
Notifications & exceptions	Customer service, Operations	SMS and email at each key step. Services down = 2 retries then save the application. Ops act on queue cases in 4 working hours, Compliance in 1 day.

Problem statement

The problem of	a slow, paper and branch-based savings account opening process with manual KYC checks
affects	customers, branch staff, central operations and the retail banking business
the impact of which is	2 to 3 days to open an account, around 30% of customers dropping off, double data entry and lost customers to digital banks
a successful solution would be	a fully digital journey on app and web with automatic KYC and AML checks, opening an account in under 10 minutes

2. Business Requirements Document (BRD)

Project	Digital Savings Account Opening
Version	1.0
Prepared by	Tushar Jain
Sponsor	Head of Retail Banking, ABC Bank
Date	10/11/2026
Status	Draft for review

Revision history

Version	Date	Author	Changes
0.1	04/11/2026	Tushar Jain	First draft
1.0	10/11/2026	Tushar Jain	Added compliance requirements and KPIs

Executive summary

Customers can only open a savings account at ABC Bank by visiting a branch, and it takes 2 to 3 working days. This project lets them do it fully online through the app and website. KYC is done through PAN check, Aadhaar OTP eKYC and Video KYC, AML screening runs automatically, and the account is created in CBS straight away. The aim is to open an account in under 10 minutes and cut branch visits for account opening by 80%.

Business objectives

ID	Objective	Target	When
BO-01	Open savings accounts fully online	Under 10 minutes end to end	3 months after launch
BO-02	Less dependency on branches	80% fewer branch visits for new accounts	6 months after launch
BO-03	More customers complete the journey	Completion rate up by 35%	6 months after launch
BO-04	Stay fully compliant	100% of accounts KYC and AML checked before activation	From launch

Scope

In scope	Out of scope
Savings accounts for resident individuals aged 18+	Current, corporate and joint accounts
Mobile app and website	NRI and minor accounts
OTP verification, PAN check, Aadhaar eKYC, Video KYC	Customers without PAN (Form 60)
Duplicate / CKYC check and AML screening	Loans, cards and other products
CBS account creation, debit card request, SMS / email	Changes to the branch process

Stakeholders

Stakeholder	Interest	Influence
Head of Retail Banking (sponsor)	More accounts at lower cost	High
Product Manager, Savings	Customer journey and targets	High

Stakeholder	Interest	Influence
Head of Compliance	KYC and Video KYC as per RBI	High
IT Head, Digital Channels	App / web build and integrations	High
AML / Fraud team	Screening and sanctions matches	Medium
CBS team	Customer and account creation in Finacle	Medium
Central Operations	Review queue for flagged cases	Medium
Video KYC officers	Running the video calls	Low
Customers	Quick, simple, paperless opening	Medium

As-Is process

Step	What happens today	Problem
1	Customer visits a branch in working hours	Hard for working people, many don't come back
2	Fills a paper form and gives PAN / Aadhaar copies	Long form, mistakes, forms sent back
3	Branch staff check originals and send the form to central ops	Courier / scan delay
4	Ops type the details into CBS	Same data entered twice
5	KYC and AML checks done manually or in a batch	1 to 2 days delay
6	Account opened, welcome kit posted	2 to 3 days in total, no updates to customer

To-Be process

Customer verifies mobile and email by OTP, enters details and PAN, uploads the PAN card (read by OCR), completes Aadhaar OTP eKYC, gets screened for AML and sanctions, does a short Video KYC call and gets the account number on screen. Only flagged cases go to a person. The full flow is in section 4.

Business requirements

ID	Requirement	Priority	Source
BR-01	Customers must be able to open a savings account online without visiting a branch	Must	Sponsor
BR-02	A normal application must take less than 10 minutes	Must	Product
BR-03	Mobile number and email must be verified by OTP	Must	IT Security
BR-04	PAN must be validated and the name matched with PAN records	Must	Compliance
BR-05	Aadhaar OTP eKYC must give verified name, DOB, address and photo	Must	Compliance
BR-06	Video KYC must be done before the account is fully active	Must	Compliance
BR-07	Duplicate customer profiles must not be created	Must	Operations
BR-08	Every applicant must go through AML and sanctions screening	Must	AML team
BR-09	CBS must create the customer ID and account straight after all checks pass	Must	CBS team
BR-10	Customers should be able to ask for a debit card	Should	Product
BR-11	Customers should get SMS / email updates at key steps	Should	Customer service
BR-12	Customers could save the application and finish later	Could	Product
BR-13	Failed or flagged applications must go to an operations review queue	Must	Operations

Compliance requirements

ID	Requirement	Source
CR-01	No account activated before KYC is complete	RBI KYC Master Direction
CR-02	Video KYC by a trained officer, live photo, original PAN on camera, location and time recorded	RBI V-CIP rules
CR-03	Screening against UN, MHA sanctions lists and internal watchlist	PMLA 2002
CR-04	KYC records uploaded to CKYC	RBI KYC Direction
CR-05	Customer consent before using Aadhaar and personal data	Aadhaar Act, DPDP Act 2023
CR-06	KYC data, Video KYC recordings and logs kept securely	PMLA record keeping
CR-07	One customer ID per customer	RBI KYC Direction

Assumptions and constraints

- Customers have a smartphone with camera, a PAN and an Aadhaar linked to their mobile.
- Video KYC officers available 9 am to 9 pm, all days.
- Launch within 6 months. CBS changes limited to existing APIs.
- Depends on NSDL, UIDAI, CKYC, AML tool, Video KYC vendor and SMS / email gateway.

Risks

ID	Risk	Impact	Mitigation
R-01	Fake or edited documents	High	Data comes from UIDAI, face match on Video KYC
R-02	PAN / Aadhaar service down	High	Retries, save application, customer continues later
R-03	Not enough Video KYC officers at peak times	Medium	Slot booking and staff planning
R-04	CBS slow or failing	High	Retry, then Ops queue, load testing
R-05	Drop-off at Video KYC step	Medium	Clear instructions, book a later slot

KPIs

KPI	Today	Target
Account opening time	2-3 working days	Under 10 minutes
Branch visits for new accounts	100%	Down 80%
Completion rate	About 70%	Up 35%
Straight-through processing (no manual step)	0%	85%
Accounts activated without KYC / AML	-	0

3. Functional Requirements Document (FRD)

Project	Digital Savings Account Opening
Version	1.0
Prepared by	Tushar Jain
Based on	BRD v1.0
Date	14/11/2026
Purpose	This explains how the system will work to meet the BRD. It covers 4 modules: onboarding, KYC & AML, CBS account creation, and notifications & exceptions.

User roles and systems

Role / system	What it does
Customer	Fills the application, uploads PAN, does eKYC and Video KYC
Video KYC officer	Runs the video call and approves, rejects or refers
Operations / Compliance user	Works on review queue cases and AML matches
NSDL, UIDAI, CKYC	PAN check, Aadhaar eKYC, central KYC records
AML tool	Sanctions, PEP and watchlist screening with a risk score
CBS (Finacle)	Creates customer ID (CIF) and savings account

Functional requirements

ID	The system shall...	BR	Priority
FR-ON-01	verify the mobile number with a 6-digit OTP (valid 3 min, max 3 tries)	BR-03	Must
FR-ON-02	verify the email with an OTP	BR-03	Must
FR-ON-03	capture name, DOB, PAN, Aadhaar, address, occupation, income and nominee (optional)	BR-01	Must
FR-ON-04	stop the journey if age is below 18	BR-01	Must
FR-ON-05	let the customer upload a PAN image (JPG, PNG, PDF, max 5 MB)	BR-04	Must
FR-ON-06	read name, DOB and PAN from the image with OCR and let the customer edit them	BR-04	Must
FR-ON-07	allow re-upload of a bad image up to 3 times, then close the application	BR-13	Must
FR-ON-08	show products by age (SAV003 only for 60+)	BR-01	Must
FR-ON-09	save the application for 7 days and send a link to continue	BR-12	Could
FR-KYC-01	validate PAN with NSDL (active, name match 80%+)	BR-04	Must
FR-KYC-02	take consent before Aadhaar eKYC and log it	CR-05	Must
FR-KYC-03	do Aadhaar OTP eKYC and store the details with Aadhaar masked	BR-05	Must
FR-KYC-04	search CKYC using PAN and pre-fill if found	CR-04	Should
FR-KYC-05	check for an existing customer in CBS by PAN and mobile	BR-07	Must
FR-KYC-06	run AML screening and act on the risk score	BR-08	Must
FR-KYC-07	put the application on hold if there is a sanctions match	CR-03	Must

ID	The system shall…	BR	Priority
FR-KYC-08	connect the customer to a Video KYC officer or let them book a slot	BR-06	Must
FR-KYC-09	record the officer's decision (approve, reject, refer) and the call	CR-02	Must
FR-CBS-01	call CBS only after PAN, eKYC, AML and Video KYC all pass	BR-09	Must
FR-CBS-02	use the existing CIF if found, otherwise create a new one	CR-07	Must
FR-CBS-03	create the savings account with the chosen product and home branch by PIN code	BR-09	Must
FR-CBS-04	show account number, IFSC and CIF on the success screen	BR-09	Must
FR-CBS-05	let the customer request a debit card	BR-10	Should
FR-CBS-06	upload the KYC record to CKYC within 10 days	CR-04	Must
FR-NT-01	send SMS / email at key steps	BR-11	Should
FR-NT-02	give a simple reason when an application fails (never the AML reason)	BR-11	Should
FR-NT-03	retry a timed-out service twice, then save the application	BR-13	Must
FR-NT-04	create an Ops queue case for failed or flagged applications	BR-13	Must
FR-NT-05	track queue SLA and escalate breaches	BR-13	Must
FR-NT-06	keep an audit log of every step	CR-06	Must

Business rules

RULE-01: Only resident individuals aged 18 or above can apply online.

RULE-02: PAN name must match the entered name by 80% or more.

RULE-03: AML risk score 0-40 continue, 41-70 continue and flag for review, 71+ stop and send to Compliance.

RULE-04: Live Video KYC 9 am to 9 pm. Outside these hours, slot booking only.

RULE-05: Customer must be in India during Video KYC (checked by location).

RULE-06: One CIF per customer. An existing customer gets the new account under the old CIF.

RULE-07: Ops queue cases closed within 4 working hours, Compliance cases within 1 working day.

RULE-08: Saved applications are deleted after 7 days.

Error messages

Code	When	Message
E-01	Wrong OTP	The OTP you entered is incorrect. Please try again.
E-02	Under 18	You must be 18 or older to open this account online.
E-03	3 failed uploads	We could not read your document. Please try again later.
E-04	PAN not verified	We could not verify your PAN details. Please check and try again.
E-05	Under review	Your application is under review. We will update you within 1 working day.
E-06	Service down	We are facing a technical issue. Your application is saved, please continue later.

Non-functional requirements

ID	Type	Requirement
NFR-01	Performance	Screens load in under 3 seconds, full journey under 10 minutes

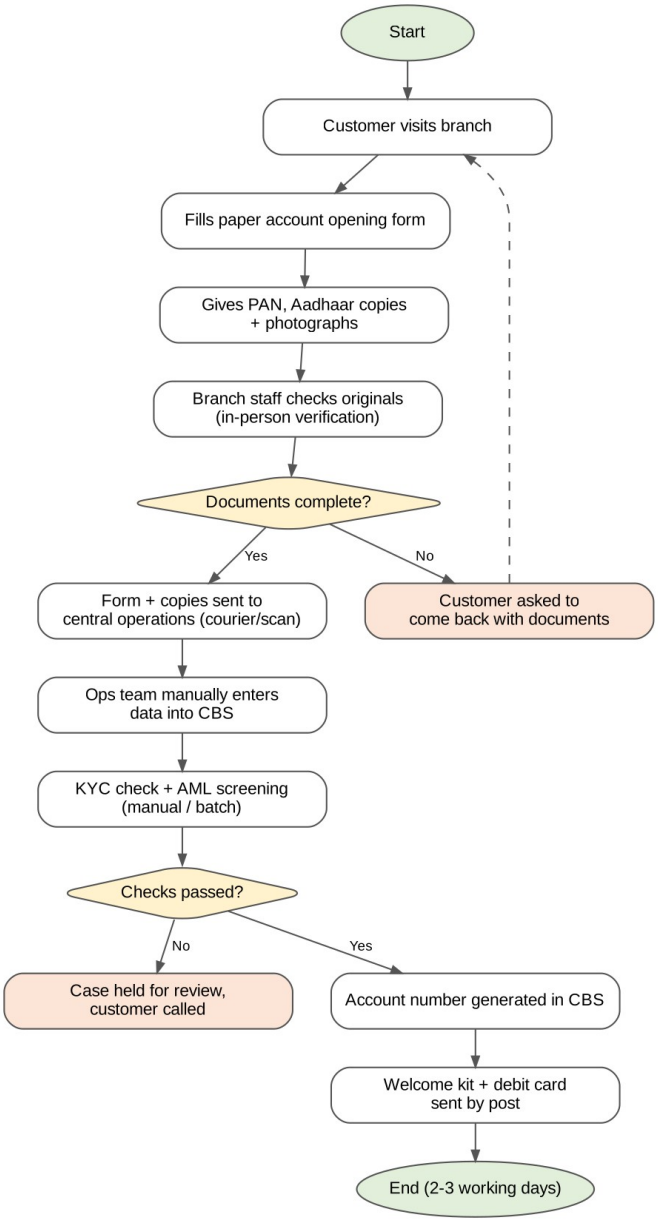
ID	Type	Requirement
NFR-02	Availability	99.5% uptime, planned downtime only 1 am - 4 am
NFR-03	Security	Data encrypted in transit and at rest, Aadhaar always masked
NFR-04	Session	Times out after 5 minutes of no activity, data saved
NFR-05	Capacity	5,000 applications a day, 200 at the same time

Traceability matrix

BR	FR	UAT
BR-01	FR-ON-03, 04, 08	TC-01, TC-03
BR-03	FR-ON-01, 02	TC-01
BR-04	FR-ON-05, 06, FR-KYC-01	TC-02, TC-04, TC-05
BR-05	FR-KYC-02, 03	TC-06
BR-06	FR-KYC-08, 09	TC-09, TC-10
BR-07	FR-KYC-05, FR-CBS-02	TC-12
BR-08	FR-KYC-06, 07	TC-07, TC-08
BR-09	FR-CBS-01, 03, 04	TC-11
BR-11	FR-NT-01, 02	TC-11, TC-13
BR-13	FR-ON-07, FR-NT-03, 04	TC-02, TC-13, TC-14

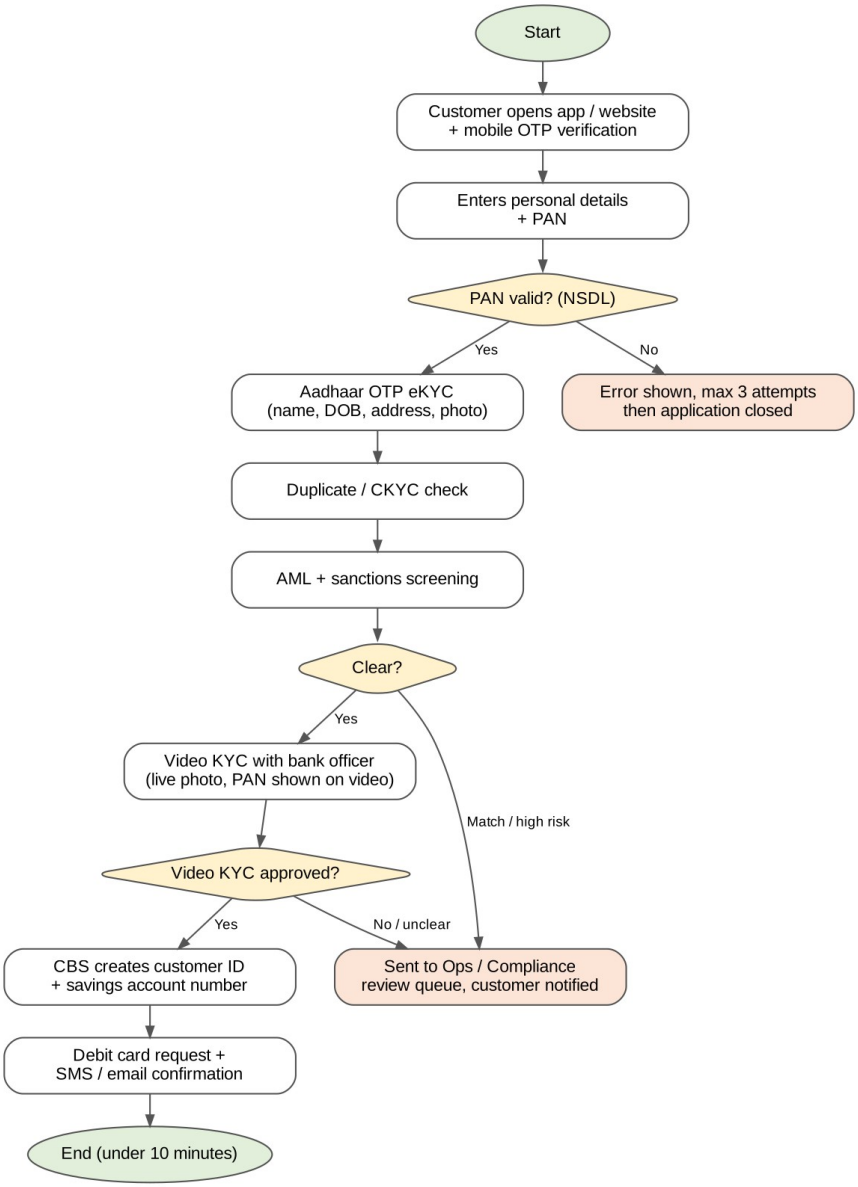
4. Process flow (As-Is / To-Be)

As-Is (branch process)



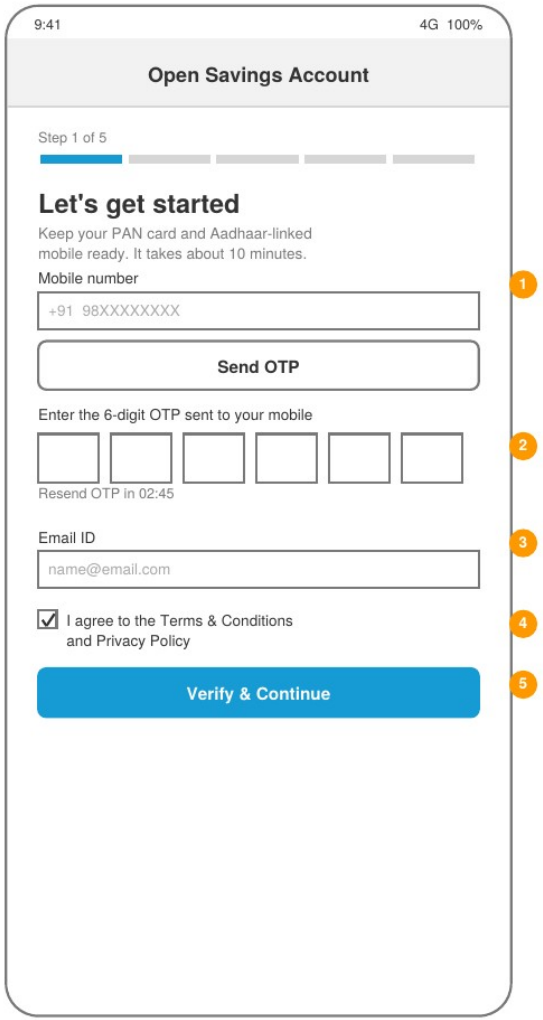
To-Be (digital process)

Orange boxes are the exception paths.



5. Wireframes

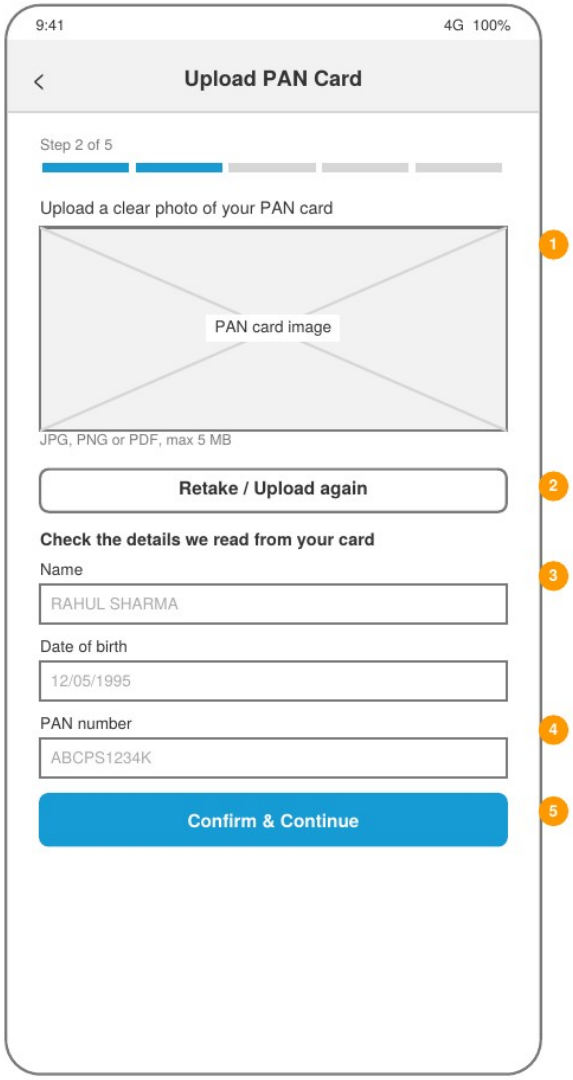
Low-fidelity screens for the mobile app. Orange numbers match the notes on each screen.



WF-01 Mobile number & OTP verification

Customer starts the journey and verifies mobile number and email.

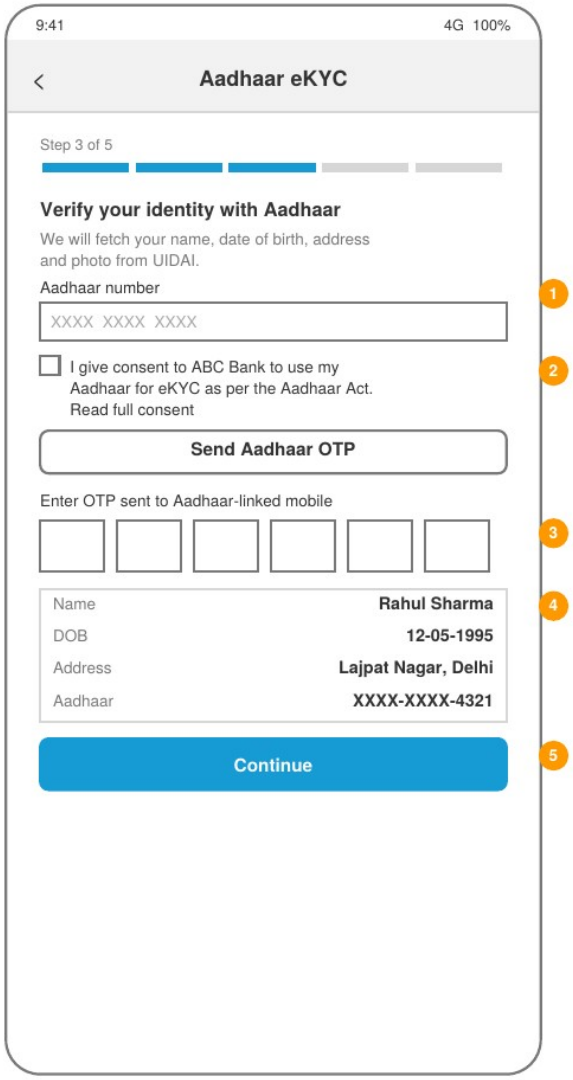
#	Element	Behaviour / business rule	Ref
1	Mobile number	10 digits, starts with 6-9. 'Send OTP' active only when number is valid.	FR-ON-01
2	OTP boxes	6-digit OTP valid for 3 minutes. Resend after timer ends. 3 wrong attempts = blocked 30 min (E-01).	FR-ON-01
3	Email ID	Valid email format. Email OTP sent on the next screen load.	FR-ON-02
4	T&C; checkbox	Must be ticked to continue. Time of acceptance logged.	FR-NT-06
5	Verify & Continue	Disabled until mobile OTP verified, email filled and T&C; ticked.	BR-03



WF-02 PAN upload & OCR review

Customer uploads PAN card. OCR reads the details and customer confirms them.

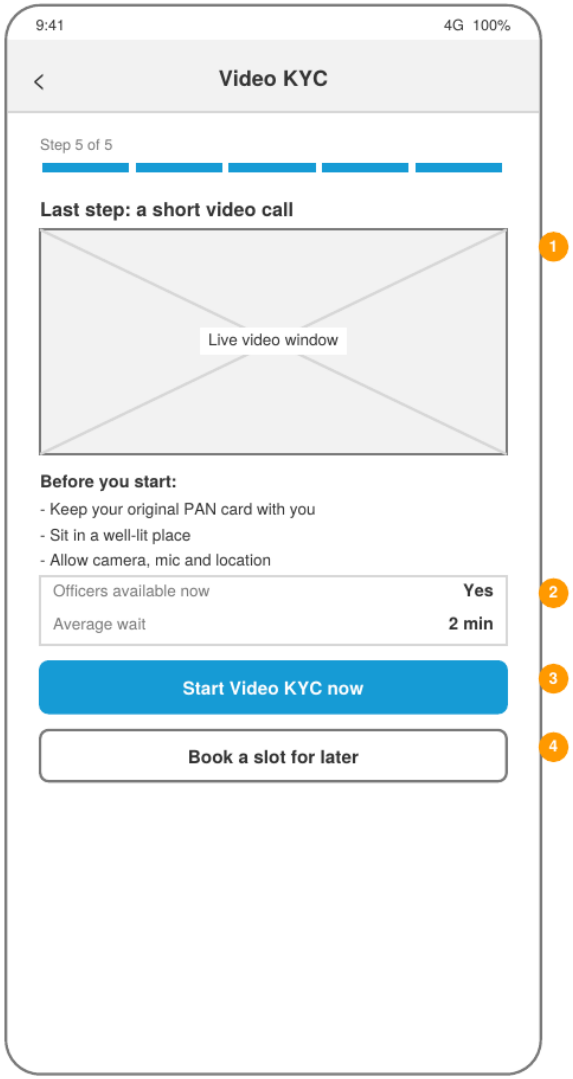
#	Element	Behaviour / business rule	Ref
1	PAN image	Camera or gallery. JPG/PNG/PDF up to 5 MB.	FR-ON-05
2	Retake / Upload again	If image is blurred or OCR confidence < 80%, message shown and re-upload asked. Max 3 attempts, then E-03.	FR-ON-06, 07
3	Extracted fields	Name, DOB, PAN filled by OCR. Customer can edit if wrong.	FR-ON-06
4	PAN number	Format check AAAAA9999A. Then NSDL check: PAN active and name match 80%+ (E-04 if fail).	FR-KYC-01
5	Confirm & Continue	Triggers PAN validation. Loader shown while waiting for NSDL.	FR-KYC-01



WF-03 Aadhaar consent & eKYC

Customer gives consent and completes Aadhaar OTP eKYC. Then duplicate, CKYC and AML checks run in background.

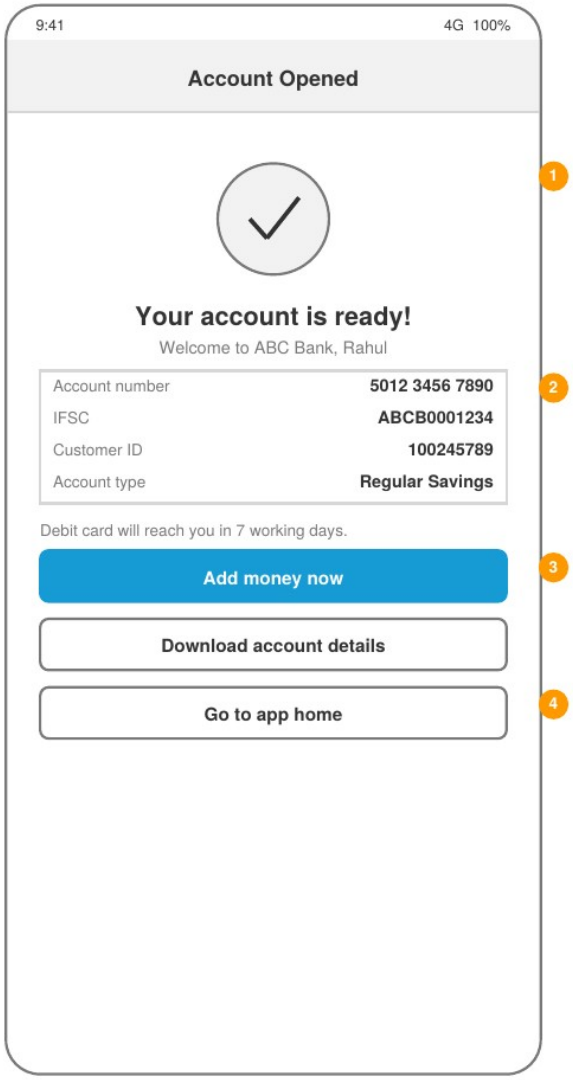
#	Element	Behaviour / business rule	Ref
1	Aadhaar number	12 digits. Masked after entry, only last 4 digits shown.	FR-KYC-03
2	Consent checkbox	Mandatory. 'Send OTP' disabled until ticked. Consent text and time logged.	FR-KYC-02
3	Aadhaar OTP	OTP from UIDAI. 3 attempts. Retry logic if UIDAI is down (E-06).	FR-KYC-03, FR-NT-03
4	eKYC details card	Read only. Data from UIDAI, cannot be edited by customer.	FR-KYC-03
5	Continue	Runs duplicate/CKYC check and AML screening. High risk or sanctions match goes to the review queue.	FR-KYC-04 to 07



WF-04 Video KYC

Customer joins a live video call with a bank officer for full KYC.

#	Element	Behaviour / business rule	Ref
1	Live video	Officer matches live face with Aadhaar photo, customer shows original PAN. Location must be in India. Call recorded with date and time.	FR-KYC-08, 09, CR-02
2	Availability	Live calls 9 am - 9 pm. Outside these hours only 'Book a slot' is shown.	FR-KYC-08
3	Start now	Opens camera after permission check. If call drops, customer can rejoin within 15 min.	FR-KYC-08
4	Book a slot	Customer picks date and time. SMS reminder sent.	FR-NT-01

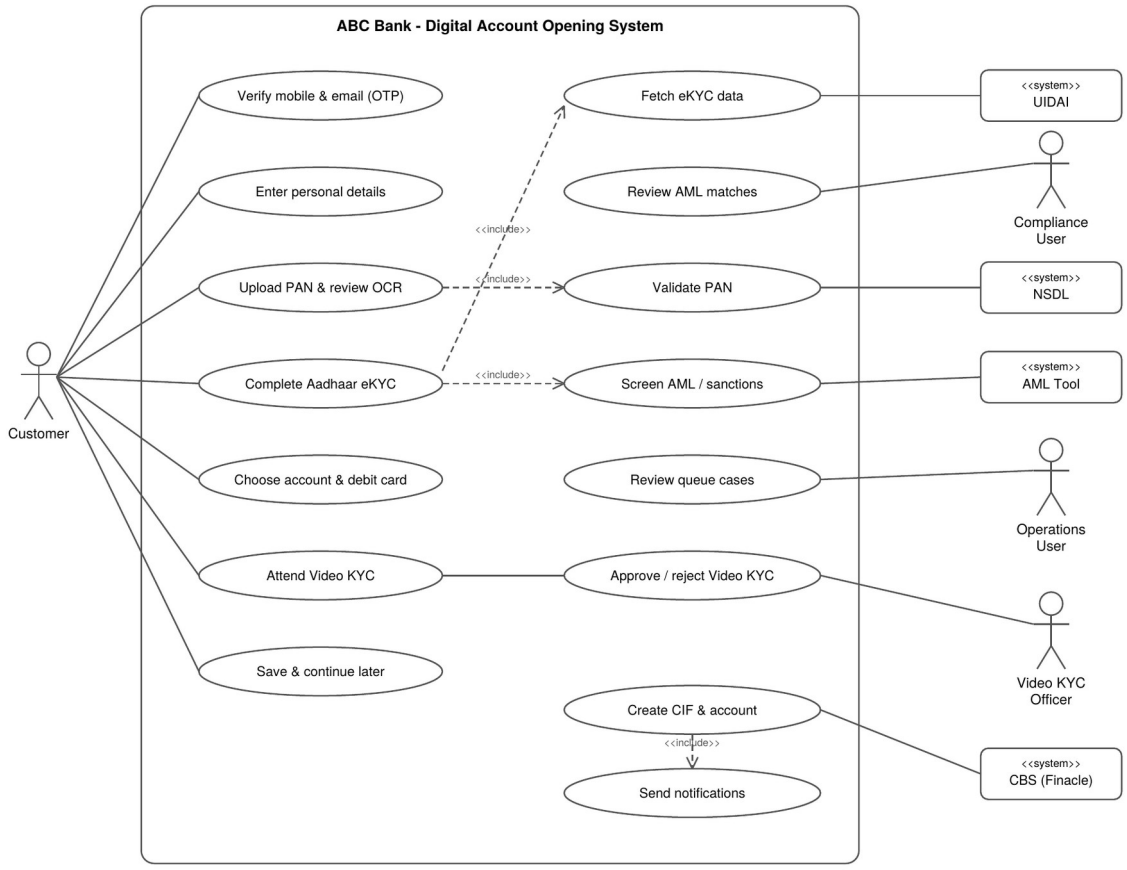


WF-05 Account opened

Success screen after CBS creates the customer ID and account.

#	Element	Behaviour / business rule	Ref
1	Success message	Shown only after CBS returns account number.	FR-CBS-01
2	Account details	Account number, IFSC and CIF shown. Same details sent by SMS and email.	FR-CBS-04, FR-NT-01
3	Add money now	Opens UPI / net banking to fund the account (existing payment flow).	Out of scope - existing
4	Go to app home	Customer is logged in to mobile banking.	-

6. Use case diagram



7. User stories

For the Jira backlog. Story points to be estimated by the dev team.

ID	Story	Acceptance criteria	FR
US-01	As a customer, I want to verify my mobile with an OTP so that my account is linked to my number	OTP valid 3 min. 3 wrong tries blocks for 30 min. Resend after timer.	FR-ON-01
US-02	As a customer, I want to upload my PAN card and have the details filled for me so that I type less	Name, DOB, PAN filled by OCR. I can edit them. Blurred image asks for re-upload.	FR-ON-05, 06
US-03	As a customer, I want to save my application and finish later so that I don't lose my progress	Saved for 7 days. SMS link opens with my data filled.	FR-ON-09
US-04	As a customer, I want to do eKYC with my Aadhaar OTP so that I don't need paper copies	Consent needed first. Name, DOB, address, photo fetched. Aadhaar shown masked.	FR-KYC-02, 03
US-05	As a compliance officer, I want every applicant screened for AML and sanctions so that we don't onboard risky customers	High risk or sanctions match goes on hold with a Compliance case and audit log.	FR-KYC-06, 07
US-06	As a customer, I want to do Video KYC from home or book a slot so that I don't visit a branch	Live calls 9 am - 9 pm. Slot booking outside these hours. Reminder SMS.	FR-KYC-08
US-07	As a Video KYC officer, I want to approve, reject or refer a call so that unclear cases get a second look	Decision saved with call recording, time and location. Refer creates Ops case.	FR-KYC-09
US-08	As an existing customer, I want my new account added to my current customer ID so that I see all accounts in one place	Match by PAN and mobile. No new CIF created.	FR-KYC-05, FR-CBS-02
US-09	As a customer, I want to see my account number as soon as checks pass so that I can start using it	Account number, IFSC and CIF on screen and by SMS / email.	FR-CBS-03, 04
US-10	As an operations user, I want failed cases in one queue with the reason so that I can fix them quickly	Case shows reason, details and documents. Escalates after 4 working hours.	FR-NT-04, 05

8. UAT test cases

Run by the product team, operations and compliance before launch (01-05 Dec 2026).

ID	Test	Steps	Expected result	FR	Result
TC-01	Successful onboarding	Verify OTP, enter details, upload valid PAN, submit	Application submitted, reference number shown	FR-ON-01, 03	Pass
TC-02	Blurred PAN image	Upload blurred image	Asked to re-upload, closed after 3 tries	FR-ON-06, 07	Pass
TC-03	Under 18	Enter DOB for age 17	E-02 shown, journey stops	FR-ON-04	Pass
TC-04	Valid PAN	Submit PAN ABCPS1234K	PAN active, name match, continue	FR-KYC-01	Pass
TC-05	Invalid PAN	Submit inactive PAN	E-04 shown	FR-KYC-01	Pass
TC-06	Aadhaar eKYC	Give consent, enter Aadhaar OTP	Details fetched, Aadhaar masked	FR-KYC-02, 03	Pass
TC-07	Low-risk AML	Customer with risk score 20	Moves to Video KYC	FR-KYC-06	Pass
TC-08	Sanctions match	Name on test sanctions list	On hold, Compliance case, audit log	FR-KYC-07	Pass
TC-09	Video KYC approve	Show PAN on call, officer approves	Approved, call recorded	FR-KYC-09	Pass
TC-10	Video KYC refer	Poor lighting, officer refers	Ops queue case, SMS sent	FR-KYC-09	Pass
TC-11	Account created	All checks pass, choose SAV001	Account number, IFSC, CIF shown, SMS sent	FR-CBS-03, 04	Pass
TC-12	Existing customer	Apply with existing customer's PAN	New account under old CIF	FR-CBS-02	Pass
TC-13	Service timeout	NSDL times out	2 retries, then saved, E-06	FR-NT-03	Pass
TC-14	Senior citizen product	Age exactly 60	SAV003 shown	FR-ON-08	Pass (retest)

Defects

ID	Test	Issue	Status
DEF-01	TC-14	SAV003 not shown at age exactly 60 (rule used 'over 60')	Fixed and retested

All 14 test cases passed (1 after retest). No open defects.